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ABSTRACT

Depression is a psychological disorder characterized by the continuous occurrence of a low mood state. It significantly affects individuals across multiple age groups worldwide and has become a serious global health concern. Early diagnosis plays a vital role in preventing its long-term impact and can potentially save lives. Electroencephalogram (EEG) signals serve as a reliable diagnostic tool, as the analysis of brain wave patterns can reflect the underlying mental state of an individual.

A fully automated depression detection system offers advantages over manual EEG analysis, which is time-consuming, labor-intensive, and requires significant expertise. A novel computer-aided hybrid neural network model, known as DepHNN (Depression Hybrid Neural Network), has been developed for depression screening using EEG signals. The model integrates a Convolutional Neural Network (CNN) for spatial and temporal feature extraction, along with Long Short-Term Memory (LSTM) layers for learning sequential patterns. The use of a windowing technique reduces computation complexity and training time. EEG data was collected from 21 drug-free symptomatic depressed patients and 24 normal individuals using a neuroscan system. The model achieved a high classification accuracy of 99.10% and a mean absolute error (MAE) of 0.2040, demonstrating its effectiveness, low complexity, and strong potential for real-time depression detection.

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CHAPTER 1

INTRODUCTION